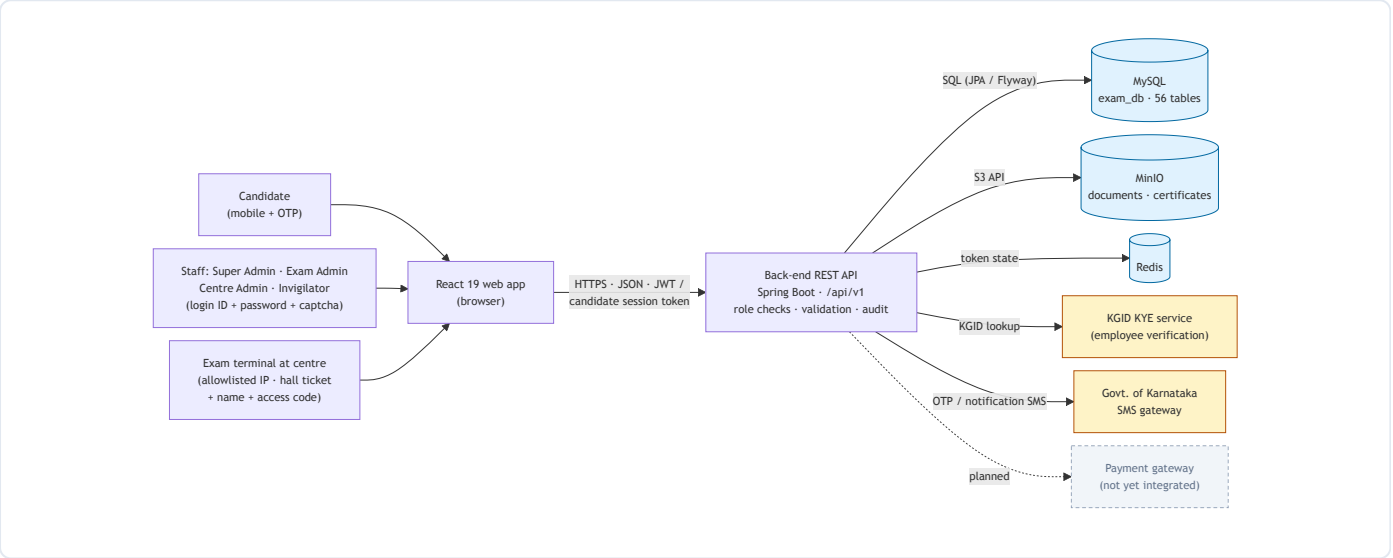
# Examination Management Portal — Data Architecture

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|---|---|
| **System** | Examination Management Portal: online registration of government employees (verified by KGID) and other-organisation candidates, examination centre management, scheduling, hall tickets, exam-day centre operations, computer-based online test, results and certificates |
| **Status** | Development / UAT (branches `main`, `release/uat`, `prod`); production deployment **to be confirmed by DevOps** |
| **Data stores** | MySQL database `exam_db` (56 tables) · MinIO object storage (uploaded documents, certificates) · Redis (login token state) |
| **Companion documents** | ExamPortal_Data_Dictionary.xlsx (tables, columns, validation rules) · ExamPortal_ER_Diagram.pdf (tables and links by area) |

## 1. End-to-end data flow




1. **Register.** A candidate logs in with mobile number and a 6-digit OTP sent by SMS (`cand_candidate`, `cand_otp_challenge`) and creates an application (`reg_application`) as a government employee (GOVT_DEPT) or other organisation (OTHER_ORG). For GOVT_DEPT the KGID number is looked up in the KGID KYE service and the returned details pre-fill the profile, which the candidate confirms (`reg_profile`). Centre preferences, exam, date and sitting are chosen.
2. **Validate and store.** The API checks the token, role and record ownership, validates every field against the rules in the data dictionary, and writes to MySQL. Every create, update and delete request, every denied access and every audit export is written to the audit trail (`audit_events`) with passwords, OTPs and tokens masked.
3. **Documents.** Photograph, ID proof and declarations are uploaded through the API (PDF / JPEG / PNG, max 5 MB) and stored in MinIO; one row per file with its object key and SHA-256 checksum is kept in `reg_document`.
4. **Fee, review and hall ticket.** The fee is assessed from the fee master (`reg_fee_assessment`, `reg_fee_master`); payment attempts are recorded in `reg_payment_attempt` (online gateway not yet integrated). After verification (`reg_review`) a hall ticket with a random QR token is generated (`reg_hall_ticket`).
5. **Exam day.** Centre staff scan the hall-ticket QR or verify identity, mark attendance and activate the test (`ops_candidate_day`, `ops_identity_verification`, `ops_attendance`, `ops_session_activation`); a 4-character access code is issued and sent by SMS (`ops_exam_portal_session`).
6. **Online test and result.** On an allowlisted terminal the candidate enters hall ticket number, name and access code. Questions are drawn per the examination's allocation (`exam_attempt_questions`), answers saved (`exam_attempt_answers`), the score calculated on submit (`exam_attempts`) and a certificate issued (`certificates`; PDF in MinIO).
7. **Notify.** OTPs, hall-ticket, exam-credential and result messages are sent through the Government of Karnataka SMS gateway using English / Kannada templates (`sms_template`); every message is logged in `sms_audit` with sensitive values such as the OTP masked.

## 2. Storage

| Store | What is stored | How |
|---|---|---|
| MySQL `exam_db` | Staff accounts, roles and permissions; masters; centres, capacity, devices and IP allowlist; examinations, question bank and schedules; candidates, applications, profiles, fees, hall tickets; exam-day records; test attempts, answers and certificates; audit trail; SMS templates and SMS log | 56 tables in nine areas; schema managed by Flyway migrations. UUID keys for staff / master data, numeric keys for registration and exam-day data. 48 foreign-key constraints and 27 check constraints |

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| MinIO object storage | Applicant documents (photograph, ID proof, declarations), document templates and generated certificate PDFs | Application bucket plus a temporary bucket; file saved under a generated object key referenced from `reg_document.file_ref`; SHA-256 checksum kept; max 5 MB per upload |
| Redis | Identifiers of issued login tokens (active / revoked) | Used for logout and token revocation; entries expire with the token (maximum 8 hours) |
| Not stored by the portal | Plain-text passwords, OTPs or captcha text; card / bank details | Passwords stored as BCrypt hashes; OTPs, captcha and session tokens only as SHA-256 hashes; no payment details (gateway not yet integrated) |

## 3. External links

| System | Purpose | Integration | Data exchanged |
|---|---|---|---|
| KGID KYE service (Karnataka Government Insurance Department) | Verify a government employee and pre-fill the profile | REST call from the back end with client authentication token; 5 s connect / 15 s read timeout. Address configured by environment variable (`KGID_MODE` must be set to the live adapter in production) | Sent: KGID number. Received: name, department, designation, e-mail, mobile, address, date of birth, date of joining, district (stored as `reg_profile.kgid_snapshot`) |
| Govt. of Karnataka SMS gateway (Mobile One) | OTPs and notifications (hall ticket, exam credentials, results) | HTTP API called from the back end with gateway account credentials supplied by environment variables; DLT template IDs | Sent: mobile number and message text. Received: message ID and gateway response (stored in `sms_audit`) |
| Payment gateway | Registration fee | **Not yet integrated**; data model ready (`reg_payment_attempt`) | — |

## 4. Retrieval and access

- **Only through the API.** The web app never connects to MySQL, MinIO or Redis directly. Staff use a signed JWT (15-minute access token, 8-hour refresh token, revocable); candidates use a session token stored only as a hash (30-minute idle, 8-hour absolute timeout).
- **Role-based access.** Endpoints declare the roles allowed: Super Admin, Examination Administrator, Examination Centre Admin, Invigilator and Candidate.
- **Record-level scope.** Centre staff see only the candidates and schedules of the centre they are bound to (`ctr_user_binding`); candidates see only their own application, documents, hall ticket and result. Documents are streamed through the API after the same check.
- **Exam portal.** The online test opens only from allowlisted IP addresses (`exam_ip_allowlist`) with hall ticket number, name and an activated access code.
- **Without login.** Open-exam list, invigilator self-registration, hall-ticket check by QR token and certificate verification by certificate code.

## 5. Protection

| Control | Implementation |
|---|---|
| In transit | HTTPS with TLS 1.3 on the public portal and API endpoint **(to be configured and confirmed by the infrastructure team)**. Note: the default SMS gateway and KGID addresses in configuration use `http://`; HTTPS or a private network link should be confirmed for production |
| Backup | Daily backups of the MySQL database and MinIO storage, kept for 30 days **(to be confirmed by the infrastructure team)** |
| Credentials | Staff passwords: 10–16 characters with upper / lower case, digit and special character, stored as BCrypt hash; account locked for 30 minutes after 5 failed attempts; login captcha. Candidate OTP: SHA-256 hash, valid 5 minutes, locked after 5 wrong attempts. Application secrets supplied through environment variables |
| Integrity and audit | Server-side validation of all inputs; foreign-key and check constraints; SHA-256 file checksums; audit trail of all changes and denied accesses with actor, IP, path, outcome and sanitised before / after data; history kept for attendance corrections, identity checks, activations and capacity changes |

Sources: examination-management-portal-backend (Spring Boot) and examination-management-portal-frontend (React) repositories, branch `main`, back-end commit 2a3186e; transport security and backup schedule to be operated and confirmed by the hosting / infrastructure team.